# Principal Component Analysis — The Physical Meaning Behind the Steps

**Data Mining (Data Reduction) — Week 6**

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## 1. The Story and the Data

In the previous session we saw a PCA example where the data obviously changed along the diagonal, so we could *see* the direction of change. Here we take a slightly different example, because the goal is to understand the **physical significance** of PCA: what it does, what it does **not** tell us, and how the **trade-off** is decided. Just like DFT, PCA has a trade-off, and following steps blindly without intuition leaves us unaware of the limitations.

We start with two variables:

$$(x_1, x_2) = (\text{hours of study},\ \text{exam score})$$

- $x_1$ = hours of study (how much a student studied)
- $x_2$ = exam score (how much they scored). The scores are **normalized**, so do not worry about how one can score "1 mark"; they are just relative numbers.

The data given to us is six points, each written as $(x_1, x_2)$:

$$(2,1),\ (3,2),\ (4,3),\ (5,3),\ (6,5),\ (7,6)$$

Reading the first point: $(2, 1)$ means **2 hours of study** and an **exam score of 1**.

As the hours go from $2$ to $7$, the score generally rises. It is not perfectly monotonic: for $4$ hours **and** for $5$ hours the score is the same $(3)$, but the overall trend is increasing. This makes physical sense (more study, higher score), although this is certainly **not true in general**; it is just a convenient example.

Each data point is a **2D vector**, and the question PCA answers is: *in which direction does this cloud of points really vary, and can one number per student replace two?*

### The roadmap

1. **Data matrix X**: A matrix of observations, where each row represents an observation and each column represents a feature. In this case, $X$ is a $6 \times 2$ matrix.
2. **Mean vector mu**: A vector containing the mean of each feature in the data matrix $X$. Here, $\mu$ is a $2 \times 1$ vector.
3. **Mean centered Xc**: The data matrix $X$ with the mean vector $\mu$ subtracted from each observation. This centers the data around the origin. $X_c = X - \mathbf{1}\mu^T$, where $\mathbf{1}$ is a $6 \times 1$ vector of ones.
4. **Covariance matrix C**: A matrix that describes the variance of each feature and the covariance between pairs of features. It's calculated as: $C = \frac{X_c^T X_c}{N-1}$, where $N$ is the number of observations (6 in this case). $C$ is a $2 \times 2$ matrix.
5. **Eigenvalues and eigenvectors**: For the covariance matrix $C$, we find its eigenvalues ($\lambda$) and corresponding eigenvectors ($v$) by solving the equation $Cv = \lambda v$.
6. **Pick PC1 = v1**: The eigenvector corresponding to the largest eigenvalue is selected as the first principal component (PC1). This eigenvector represents the direction of maximum variance in the data. $v_1$ is a $2 \times 1$ vector.
7. **Reduce: Z1 = Xc v1**: The mean-centered data $X_c$ is projected onto the first principal component $v_1$. This reduces the dimensionality of the data. $Z_1$ is a $6 \times 1$ vector.
8. **Estimate: X_hat = Z1 v1^T + mu**: The original data is reconstructed using the projected data $Z_1$, the first principal component $v_1$, and the mean vector $\mu$. This is an approximation of the original data $X$. $\hat{X} = Z_1 v_1^T + \mathbf{1}\mu^T$. $\hat{X}$ is a $6 \times 2$ matrix.

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## 2. Step 1: Build the Data Matrix

Arrange the data into a matrix called the **data matrix** $X$. Each row is a data point, each column is a variable (feature).

$$X = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 3 \\ 5 & 3 \\ 6 & 5 \\ 7 & 6 \end{bmatrix}$$

- The first column is **hours of study** ($x_1$).
- The second column is **exam score** ($x_2$).
- There are $N = 6$ data points, and remember each data point is a 2D vector (for example $(2, 1)$ is the first data point and $(3, 2)$ is the second).

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## 3. Step 2: Compute the Mean Vector

Define the **mean vector** $\mu$. With two variables, it has two entries.

The two columns have very different physical meaning: one is in hours, the other is just (unitless) marks. You obviously do **not** add marks to hours, so the mean must be computed **within each variable (column)**, never across the two variables.

$$\mu_1 = \frac{2+3+4+5+6+7}{6} = \frac{27}{6} = 4.5$$

$$\mu_2 = \frac{1+2+3+3+5+6}{6} = \frac{20}{6} \approx 3.33$$

$$\mu = \begin{bmatrix} 4.5 \\ 3.33 \end{bmatrix}$$

> **Note for Review:** Do **not** take the mean of $2$ and $1$ (the entries of one row). Those are different physical quantities. Compute $\mu$ only *within* a variable: the mean of $x_1$ for $x_1$, the mean of $x_2$ for $x_2$, and if there are more variables, one mean per variable. Never across them.

## 4. Step 3: Mean Centering

**Mean centering** means subtracting the mean from each data point. The new centered data matrix is denoted $X_c$ (the subscript $c$ says the data is centered around the mean).

For each row $X_r$ of $X$ (a row is one data point), compute:

$$X_c \text{ row} = X_r - \mu$$

For example, the first point is $(2, 1)$, so $(2 - 4.5,\ 1 - 3.33) = (-2.5,\ -2.33)$. Doing this for every row:

| Point | $x_1 - 4.5$ | $x_2 - 3.33$ |
|---|---|---|
| $(2, 1)$ | $2 - 4.5 = -2.5$ | $1 - 3.33 = -2.33$ |
| $(3, 2)$ | $3 - 4.5 = -1.5$ | $2 - 3.33 = -1.33$ |
| $(4, 3)$ | $4 - 4.5 = -0.5$ | $3 - 3.33 = -0.33$ |
| $(5, 3)$ | $5 - 4.5 = 0.5$ | $3 - 3.33 = -0.33$ |
| $(6, 5)$ | $6 - 4.5 = 1.5$ | $5 - 3.33 = 1.67$ |
| $(7, 6)$ | $7 - 4.5 = 2.5$ | $6 - 3.33 = 2.67$ |

$$X_c = \begin{bmatrix} -2.5 & -2.33 \\ -1.5 & -1.33 \\ -0.5 & -0.33 \\ 0.5 & -0.33 \\ 1.5 & 1.66 \\ 2.5 & 2.66 \end{bmatrix} = [f_1 \quad f_2]$$

(The last two entries of the second column are written as $1.66$ and $2.66$ in the lecture working; the exact values are $\frac{5}{3}$ and $\frac{8}{3}$, i.e. $1.67$ and $2.67$ when rounded. The difference is only rounding.)

The columns of $X_c$ are the **feature vectors**:

- $f_1$ = the mean-centered **hours of study** (first column)
- $f_2$ = the mean-centered **exam score** (second column)

So the data now comes arranged as one feature vector per column.

> **Note for Review:** A quick sanity check for centering: every column of $X_c$ must sum to $0$. For $f_1$: $-2.5 - 1.5 - 0.5 + 0.5 + 1.5 + 2.5 = 0$ ✓. For $f_2$ (using exact thirds): $-\frac{7}{3} - \frac{4}{3} - \frac{1}{3} - \frac{1}{3} + \frac{5}{3} + \frac{8}{3} = 0$ ✓.

## 5. Step 4: The Covariance Matrix

This step is **new** compared with the previous example. Earlier the data visibly varied along the diagonal $(1, 1)$, so the direction of change was obvious by eye. In general you will **not** be able to see the direction of change just by looking at the data, so we need a general method: compute the **covariance matrix**, then find its eigenvalues and eigenvectors, which give the principal components.

The covariance matrix is defined as:

$$C = \frac{1}{N-1} X_c^T X_c$$

Here $N = 6$, so $N - 1 = 5$ (in general $N$ can be any number of data points):

$$C = \frac{1}{5} X_c^T X_c$$

### Dimensions

- $X_c$ is $6 \times 2$ (6 rows, 2 columns)
- $X_c^T$ is $2 \times 6$
- So $X_c^T X_c$ is $(2 \times 6)(6 \times 2) = 2 \times 2$, which has **4 numbers**

The size of $C$ depends on how many variables you started with: **2 variables give a** $2 \times 2$ **matrix; 5 variables would give a** $5 \times 5$ **matrix.**

## What sits inside $C$

$$C = \begin{bmatrix} \mathrm{Var}(f_1) & \mathrm{Cov}(f_1, f_2) \\ \mathrm{Cov}(f_1, f_2) & \mathrm{Var}(f_2) \end{bmatrix}_{2\times 2}$$

- Diagonal entries: the **variance** of each feature vector.
- Off-diagonal entries: the **covariance** between the two feature vectors (the same number appears twice, so $C$ is symmetric).

Nothing here is a brand-new quantity: we already know variance and dot products. They simply come together in one special matrix called the **covariance matrix**.

## The variances

We know how to compute variance: take $(x_i - \mu)^2$, sum, and divide by $N - 1$. Since $X_c$ is already centered, this is just the sum of squares of each column:

$$\mathrm{Var}(f_1) = \frac{1}{5}\left[(-2.5)^2 + (-1.5)^2 + (-0.5)^2 + 0.5^2 + 1.5^2 + 2.5^2\right]$$

$$= \frac{6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25}{5} = \frac{17.5}{5} = 3.5$$

$$\mathrm{Var}(f_2) = \frac{1}{5}\left[(-2.33)^2 + (-1.33)^2 + (-0.33)^2 + (-0.33)^2 + 1.67^2 + 2.67^2\right]$$

$$\approx \frac{5.44 + 1.78 + 0.11 + 0.11 + 2.78 + 7.11}{5} = \frac{17.33}{5} \approx 3.47$$

## The covariance

**Covariance is just a dot product.** For two vectors $f_1$ and $f_2$:

$$\mathrm{Cov}(f_1, f_2) = \frac{1}{N-1}\sum_{i=1}^{N} f_{1,i} f_{2,i}$$

The sum $\sum f_{1,i} f_{2,i}$ is exactly the **dot product** of $f_1$ and $f_2$, and we scale it by $\frac{1}{N-1}$ (here, divide by 5).

Computing the dot product term by term:

| $i$ | $f_{1,i}$ | $f_{2,i}$ | $f_{1,i} f_{2,i}$ |
|---|---|---|---|
| 1 | $-2.5$ | $-2.33$ | 5.83 |
| 2 | $-1.5$ | $-1.33$ | 2.00 |
| 3 | $-0.5$ | $-0.33$ | 0.17 |
| 4 | 0.5 | $-0.33$ | $-0.17$ |
| 5 | 1.5 | 1.67 | 2.50 |
| 6 | 2.5 | 2.67 | 6.67 |
| | | **Sum** | **17.0** |

$$\mathrm{Cov}(f_1, f_2) = \frac{17}{5} = 3.4$$

### The full matrix multiplication (all four entries at once)

The same numbers appear if we multiply $X_c^T X_c$ out in full:

$$X_c^T X_c = \begin{bmatrix} -2.5 & -1.5 & -0.5 & 0.5 & 1.5 & 2.5 \\ -2.33 & -1.33 & -0.33 & -0.33 & 1.67 & 2.67 \end{bmatrix} \begin{bmatrix} -2.5 & -2.33 \\ -1.5 & -1.33 \\ -0.5 & -0.33 \\ 0.5 & -0.33 \\ 1.5 & 1.67 \\ 2.5 & 2.67 \end{bmatrix}$$

Each entry is (row of $X_c^T$) · (column of $X_c$):

$$(1,1) = f_1 \cdot f_1 = 6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25 = 17.5$$
$$(1,2) = f_1 \cdot f_2 = 5.83 + 2.00 + 0.17 - 0.17 + 2.50 + 6.67 = 17.0$$
$$(2,1) = f_2 \cdot f_1 = 17.0$$
$$(2,2) = f_2 \cdot f_2 = 5.44 + 1.78 + 0.11 + 0.11 + 2.78 + 7.11 \approx 17.33$$

$$X_c^T X_c = \begin{bmatrix} 17.5 & 17.0 \\ 17.0 & 17.33 \end{bmatrix} \quad \Rightarrow \quad C = \frac{1}{5} \begin{bmatrix} 17.5 & 17.0 \\ 17.0 & 17.33 \end{bmatrix}$$

### The covariance matrix

$$C = \begin{bmatrix} 3.5 & 3.4 \\ 3.4 & 3.47 \end{bmatrix}$$

> **Note for Review:** The covariance matrix is $2 \times 2$ because we have **two variables**, not because we have two data points. The large off-diagonal entry ($3.4$, almost as large as the variances themselves) is the mathematical footprint of "hours and score rise together", and it is exactly this correlation that PCA will exploit.

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## 6. Step 5: Eigenvalues and Eigenvectors

You already know eigenvalues and eigenvectors, so only the use is stated here. Use the covariance matrix $C$ to find the eigenvector $v$ and eigenvalue $\lambda$:

$$C\,v = \lambda\,v$$

- $v$ is the **eigenvector**
- $\lambda$ is the **eigenvalue**
- There can be more than one eigenvalue, and correspondingly that many eigenvectors.

The standard way to solve it:

$$\det(C - \lambda I) = 0$$

For our $C$:

$$\det \begin{bmatrix} 3.5 - \lambda & 3.4 \\ 3.4 & 3.47 - \lambda \end{bmatrix} = (3.5 - \lambda)(3.47 - \lambda) - 3.4^2 = 0$$

This is a **quadratic equation in** $\lambda$. Expanding gives $\lambda^2 - (\text{trace})\lambda + \det(C) = 0$, where the trace is $3.5 + 3.4667 = 6.9667$ and $\det(C) = 3.5 \times 3.4667 - 3.4^2 = 0.5733$ (using unrounded entries):

$$\lambda = \frac{6.9667 \pm \sqrt{6.9667^2 - 4(0.5733)}}{2} = \frac{6.9667 \pm \sqrt{46.24}}{2} = \frac{6.9667 \pm 6.80}{2}$$

$$\boxed{\lambda_1 \approx 6.88, \qquad \lambda_2 \approx 0.083}$$

There are two eigenvalues because the covariance matrix is $2 \times 2$, and each eigenvalue has a corresponding eigenvector (found in Step 6). Before computing the eigenvectors, let us understand what these two numbers *mean*.

## 7. What Do the Eigenvalues Mean Physically

**Eigenvalues of the covariance matrix represent the amount of variance in each direction.**

Recall that PCA is about **direction**: we want the direction in which the data is varying. Now we can be more specific:

- The **eigenvalues** tell us *how much* the data varies in a particular direction.
- The **eigenvectors** (found next) tell us *which direction* that is. The first eigenvector is the direction of **maximum variance**; along the second eigenvector the data varies a bit less, and so on.

Variance is essentially **change**, and change is what we care about: it is the change that distinguishes student A from student B.

### Total variation

Since $\lambda_1$ and $\lambda_2$ are the variances along two different directions (two directions because there are two variables), the **total variation** in the data is:

$$\text{Total variation} = \lambda_1 + \lambda_2 = 6.88 + 0.083 \approx 6.966$$

This number does not have a clear physical meaning on its own, but it is useful for **comparison**: given two or more datasets, you can tell which one varies more and which one varies less.

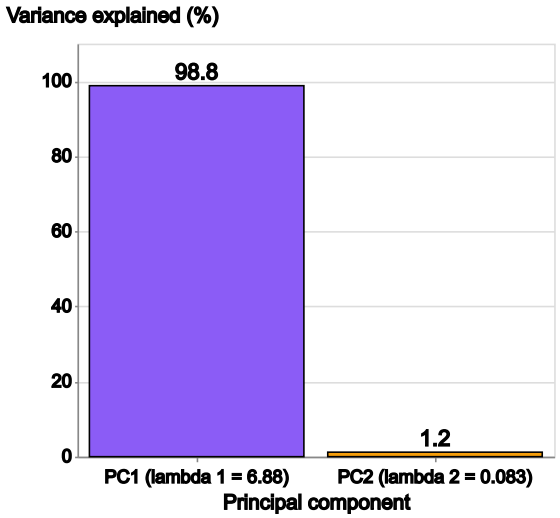
> **Note for Review:** Total variation equals the **trace** of $C$ (sum of diagonal entries): $3.5 + 3.47 = 6.97$. Rotating the axes to the eigen-directions does not create or destroy variance; it only **redistributes** it, piling it almost entirely onto the first direction.

### Percentage of variance explained

Name the directions **PC1** and **PC2** (principal component 1 and 2), the directions of $\lambda_1$ and $\lambda_2$ respectively.

$$\text{Variance explained by PC1} = \frac{\lambda_1}{\lambda_1 + \lambda_2} \times 100 = \frac{6.88}{6.966} \times 100 \approx 98.8\%$$

$$\text{Variance explained by PC2} = \frac{\lambda_2}{\lambda_1 + \lambda_2} \times 100 = \frac{0.083}{6.966} \times 100 \approx 1.2\%$$




So the direction that corresponds to $\lambda_1$ (PC1) carries **98.8%** of all the variation in the data, which is almost everything, and nearly nothing is left for the second direction. Out of the data we have found *one* direction that holds almost all the change.

With more variables, the variance spreads over more components but always in **decreasing** order. For example, the first component might hold $90\%$, the second $7.9\%$, the third $2.1\%$. **The first principal component always corresponds to the largest eigenvalue and always captures the most variation in the data.** That is how principal components work.

## 8. Step 6: Finding the Direction of Variance

The eigenvalues told us *how much* variance exists ($6.88$ along one direction and $0.083$ along the other). Now we want the actual **directions**, and those are the **eigenvectors**.

Use the same equation:

$$C\,v_1 = \lambda_1\,v_1 = 6.88\,v_1$$

To solve for $v_1 = (a, b)^T$, rewrite as $(C - 6.88\,I)v_1 = 0$:

$$\begin{bmatrix} 3.5 - 6.88 & 3.4 \\ 3.4 & 3.47 - 6.88 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -3.38 & 3.4 \\ 3.4 & -3.41 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The first row gives $-3.38\,a + 3.4\,b = 0$, so $b \approx 0.994\,a \approx a$. The direction is therefore $(1, 1)$. Normalizing to unit length (divide by $\sqrt{1^2 + 1^2} = \sqrt{2}$):

$$v_1 \approx \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \text{PC1}$$

Likewise for the second eigenvalue $\lambda_2 = 0.083$ (the same process gives $3.42\,a + 3.4\,b = 0$, so $b \approx -a$):

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \text{PC2}$$

**Check of perpendicularity:** $v_1 \cdot v_2 = \frac{1}{2}[(1)(-1) + (1)(1)] = 0$, so $v_1 \perp v_2$.

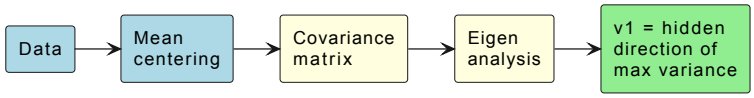
Eigenvectors of a covariance matrix are always **perpendicular** to each other. With three eigenvectors, $v_1 \perp v_2$, $v_1 \perp v_3$ and $v_2 \perp v_3$. This is a property that holds by definition.

### Interpreting the result

- $v_1$ (PC1) is the **eigenvector corresponding to $\lambda_1$, the direction of maximum variance**. Along it lies 98.8% of the variation.
- $v_2$ (PC2) is the direction holding the remaining variation (1.2%).

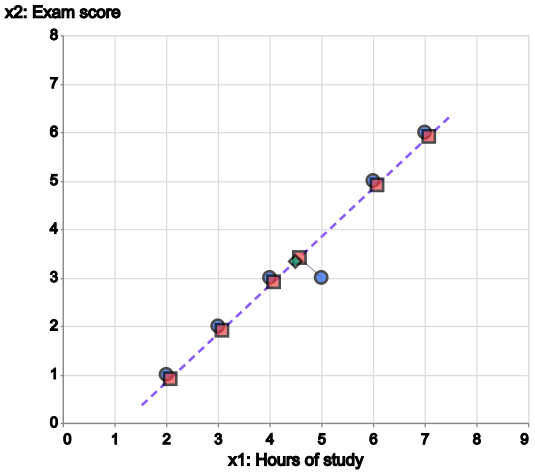
This is the **same vector** we established in the previous example, where we just *looked* at the data and saw that it varied along the diagonal. The new thing is that **we did not visualize anything**. We only used the data: mean-centered it, built the covariance matrix, did the eigenvalue and eigenvector analysis, and the direction fell out.

The data lies somewhere in 2D space and varies in a certain direction that, just by looking, may be hidden. PCA's job is to **bring that hidden dimension out** through this process:




### Seeing it on a graph

The dashed line is PC1: it passes through the mean $(4.5, 3.33)$ in the direction $\frac{1}{\sqrt{2}}(1, 1)$. Notice how closely the six points hug it. The red points and thin grey lines show what happens in Steps 7 and 8 when everything is squeezed onto that single line.




> **Note for Review:** Sign of an eigenvector is arbitrary: $v$ and $-v$ are both valid eigenvectors for the same $\lambda$. If your calculator returns $(-0.707, -0.707)$ instead of $(0.707, 0.707)$, nothing is wrong. Also, the exact eigenvector is $(0.709, 0.705)$; it rounds to the clean $\frac{1}{\sqrt{2}}(1, 1)$ because the two variances ($3.5$ and $3.47$) are almost equal.

## 9. Step 7: Data Reduction

We have now reached the actual **data reduction**. All the previous steps (centering, covariance, eigenvalues, eigenvectors) were preparation.

Both $v_1$ and PC1 represent the **direction of maximum variance**, and it accounts for more than $98\%$ of the variation. Define a new vector $Z_1$ by projecting the centered data onto $v_1$:

$$Z_1 = X_c\, v_1$$

### Dimensions

- $X_c$ is $6 \times 2$
- $v_1$ is $2 \times 1$ (one column, two rows)
- So $Z_1$ is $6 \times 1$: **one column with six elements**

### The multiplication, row by row

$$Z_1 = \begin{bmatrix} -2.5 & -2.33 \\ -1.5 & -1.33 \\ -0.5 & -0.33 \\ 0.5 & -0.33 \\ 1.5 & 1.67 \\ 2.5 & 2.67 \end{bmatrix} \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix} = \begin{bmatrix} (-2.5)(0.707) + (-2.33)(0.707) \\ (-1.5)(0.707) + (-1.33)(0.707) \\ (-0.5)(0.707) + (-0.33)(0.707) \\ (0.5)(0.707) + (-0.33)(0.707) \\ (1.5)(0.707) + (1.67)(0.707) \\ (2.5)(0.707) + (2.67)(0.707) \end{bmatrix}$$

Since both entries of $v_1$ are equal, each row simplifies to $0.707 \times (x_{1c} + x_{2c})$:

| Row | $x_{1c} + x_{2c}$ | $\times 0.707$ |
|---|---|---|
| 1 | $-2.5 - 2.33 = -4.83$ | $-3.418$ |
| 2 | $-1.5 - 1.33 = -2.83$ | $-2.004$ |
| 3 | $-0.5 - 0.33 = -0.83$ | $-0.590$ |
| 4 | $0.5 - 0.33 = 0.17$ | $0.119$ |
| 5 | $1.5 + 1.67 = 3.17$ | $2.239$ |
| 6 | $2.5 + 2.67 = 5.17$ | $3.653$ |

$$Z_1 = \begin{bmatrix} -3.418 \\ -2.004 \\ -0.590 \\ 0.119 \\ 2.239 \\ 3.653 \end{bmatrix}_{6\times 1}$$

(These are approximate numbers; the last digit can differ slightly depending on how much the intermediate values were rounded.)

$Z_1$ is the **projection of the data ($X_c$) onto the first eigenvector** $v_1$. You could likewise project onto $v_2$ and get another column, $Z_2$.

### The reduction itself

Since $v_1$ already accounts for more than $98\%$ of the variation, we do not worry about the remaining $1.2\%$. We simply **project onto $v_1$ and forget the other part**. This is very similar to DFT: we trade off some accuracy for data reduction.

| | **Original $X$** | **After PCA $Z_1$** |
|---|---|---|
| Shape | $6 \times 2$ | $6 \times 1$ |
| Numbers to store | 12 | 6 |

Instead of storing the $12$ numbers of the original matrix

$$X = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 3 \\ 5 & 3 \\ 6 & 5 \\ 7 & 6 \end{bmatrix} \quad \Longrightarrow \quad Z_1 = \begin{bmatrix} -3.418 \\ -2.004 \\ -0.590 \\ 0.119 \\ 2.239 \\ 3.653 \end{bmatrix}$$

we store just these **6 numbers**. This is similar to DFT, where we stored only $X(0)$ and $X(1)$ and obtained a reduction in data. We lose some accuracy, but the saving is significant.

> **Note for Review:** To be strictly honest about storage, reconstruction also needs $v_1$ (2 numbers) and $\mu$ (2 numbers), so the true cost here is $6 + 2 + 2 = 10$ numbers, only slightly below 12. For realistic data with thousands of rows, those few extra numbers are negligible and the saving approaches the full $2\times$. In a dataset with $d$ variables keeping $k$ components, the storage drops from $N \cdot d$ to about $N \cdot k$.

## 10. Step 8: Data Estimation

The final step is the reverse of the previous one. The numbers in $Z_1$ (some negative, some positive) are just internal working: what we ultimately want is our **original data**. So we need to **get** $X$ **back** from $Z_1$. This is **data estimation (reconstruction)**.

$$\hat{X} = X_c\, v_1\, v_1^T + \mu = Z_1\, v_1^T + \mu$$

- $X_c v_1$ is just $Z_1$.
- $v_1^T$ is the transpose of the two-dimensional vector $v_1$, i.e. $[0.707,\ 0.707]$. Shape check: $Z_1$ is $6 \times 1$, $v_1^T$ is $1 \times 2$, so $Z_1 v_1^T$ is $6 \times 2$ again.
- $\mu$ is added back, because we **subtracted the mean** during mean centering. To get the data back, we must add it back.

### The calculation, row by row

Each row is $z_i \cdot [0.707,\ 0.707] + [4.5,\ 3.33]$:

| **Row** | $z_i$ | $z_i \times 0.707$ | $+\ 4.5$ **(first column)** | $+\ 3.33$ **(second column)** |
|---|---|---|---|---|
| 1 | $-3.418$ | $-2.417$ | 2.08 | 0.91 |
| 2 | $-2.004$ | $-1.417$ | 3.08 | 1.91 |
| 3 | $-0.590$ | $-0.417$ | 4.08 | 2.91 |
| 4 | 0.119 | 0.084 | 4.58 | 3.41 |
| 5 | 2.239 | 1.583 | 6.08 | 4.91 |
| 6 | 3.653 | 2.583 | 7.08 | 5.91 |

$$\hat{X} = \begin{bmatrix} 2.08 & 0.91 \\ 3.08 & 1.91 \\ 4.08 & 2.91 \\ 4.58 & 3.41 \\ 6.08 & 4.91 \\ 7.08 & 5.91 \end{bmatrix}$$

This is the **reconstructed data**.

### Comparing with the original

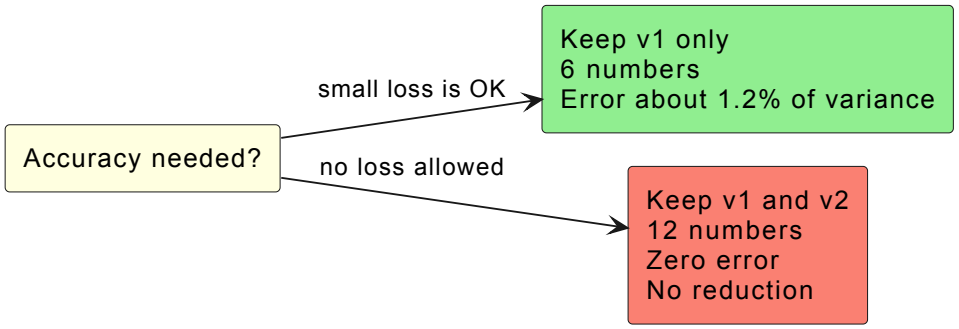
| **Case** | **Original** $(x_1, x_2)$ | **Reconstructed** $(\hat{x}_1, \hat{x}_2)$ | **Error** $(\hat{x}_1 - x_1,\ \hat{x}_2 - x_2)$ |
|---|---|---|---|
| 1 | $(2, 1)$ | $(2.08, 0.91)$ | $(+0.08, -0.09)$ |
| 2 | $(3, 2)$ | $(3.08, 1.91)$ | $(+0.08, -0.09)$ |
| 3 | $(4, 3)$ | $(4.08, 2.91)$ | $(+0.08, -0.09)$ |
| 4 | $(5, 3)$ | $(4.58, 3.41)$ | $(-0.42, +0.41)$ |
| 5 | $(6, 5)$ | $(6.08, 4.91)$ | $(+0.08, -0.09)$ |
| 6 | $(7, 6)$ | $(7.08, 5.91)$ | $(+0.08, -0.09)$ |

For instance, we get $7.08$ instead of $7$, $5.91$ instead of $6$, $3.08$ instead of $3$ and $1.91$ where we wanted $2$. Once again, just like DFT, we get **some inaccuracies**, and that is why there is a **trade-off**.

> **Note for Review:** Look at the error pattern. Five points are off by a nearly identical tiny shift, while the point $(5, 3)$ takes most of the error. That is the point that sits *off the diagonal*, the same one that breaks the "score keeps rising" pattern (the score stays at $3$ for both $4$ and $5$ hours). PCA with one component forces every point onto the PC1 line, so whatever lies away from that line (the PC2 part of the data) is what gets erased. The total squared error is $\sum \|x - \hat{x}\|^2 \approx 0.417$, which equals $(N-1)\lambda_2 = 5 \times 0.083$. The discarded eigenvalue is literally the price you pay.

### How to reduce the error

If we wanted to be more accurate, the only option is to take the second eigenvector $v_2$ as well, i.e. project onto $v_2$ too, and then we would get the entire data back exactly. But that makes no sense, because we would have the **same number of values before and after** the reduction. In practice, for large datasets small errors of $2\%$, $5\%$ or $10\%$ often do not matter. The moment you admit some inaccuracy or approximation, PCA and DFT come into the picture.




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## 11. Lossy Methods and the Trade-Off

Both **PCA and DFT are lossy methods for data reduction**. "Lossy" means some information is lost, and that is where the **trade-off** comes from.

- Be careful about **where** you apply them. For example, with **medical data** you had better not lose information, so you should not apply this kind of reduction there.
- If you are comfortable with a trade-off, you can use these methods.

There are several such methods:

| Method | Full name | Remark |
|---|---|---|
| **PCA** | Principal Component Analysis | Very popular and interesting; eigenvalue and eigenvector based |
| **DFT** | Discrete Fourier Transform | Complex exponential based |
| **DWT** | Discrete Wavelet Transform | Slightly better than DFT, but quite mathematical and it deviates from the main topic of data mining; it is mentioned in the textbook but not covered in depth here |
| **DCT** | Discrete Cosine Transform | Another member of the family |

PCA and DFT are used here as **examples** of lossy data-reduction methods. The key message is not the list of names but that you must be careful about when you apply them.

## 12. PCA versus DFT

The two methods operate in very different ways, yet at a closer look share a lot.

### Different on the surface

| | PCA | DFT |
|---|---|---|
| Mathematical machinery | **Eigenvector / eigenvalue** based | **Complex exponentials** |
| Basis functions | **Eigenvectors**, derived *from the data itself* (from the eigen-decomposition of the covariance matrix) | **Complex exponentials**, $e^{i2\pi nk/N}$, fixed in advance |
| What is measured | **Change / variation** (an eigenvalue is the change along one direction) | **Frequency** |

### Same underneath

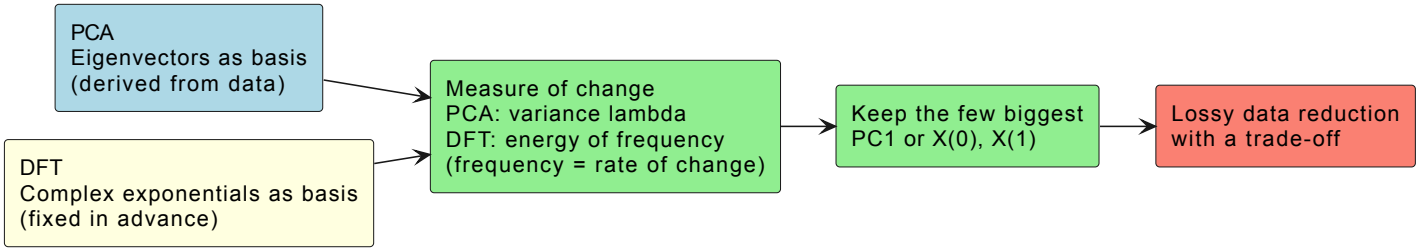
- PCA is about **change or variation** (the direction of change).
- DFT is about **frequency**, and **frequency is simply the rate of change**. For example, a $2\,\mathrm{Hz}$ sine means $2$ cycles per second, and a $20\,\mathrm{Hz}$ sine means $20$ cycles per second.

So even though they come from different principles, both exploit the same underlying property: **change**. One does it through eigenvalues along directions, the other through complex basis functions and the idea of frequency.

### How each one reduces data

| | PCA | DFT |
|---|---|---|
| What you rank by | Eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \ldots$ (variance) | Magnitude of coefficients (energy) |
| What you keep | Only the first (few) eigenvectors, e.g. only $\lambda_1$ and $v_1$ (PC1) | Only the highest-energy coefficients, e.g. $X(0), X(1)$ |
| What you discard | The remaining eigenvalues and eigenvectors | The low-energy coefficients |

In both cases you **choose directions or coefficients based on energy, magnitude or variance**, which in PCA is $\lambda$. You take the first few and discard the rest. In our example we kept only $\lambda_1$ and $v_1$; in DFT we kept only $X(0)$ and $X(1)$, because the other energies were relatively small.




> **Note for Review:** Exams like the question "how are PCA and DFT different and how are they similar?" Differ in **basis** (data-derived eigenvectors vs fixed complex exponentials). Similar in **philosophy** (rank components by variance or energy, keep the biggest, accept loss). The basis functions also have interesting mathematical properties beyond physical significance, but these are outside the scope of this course.

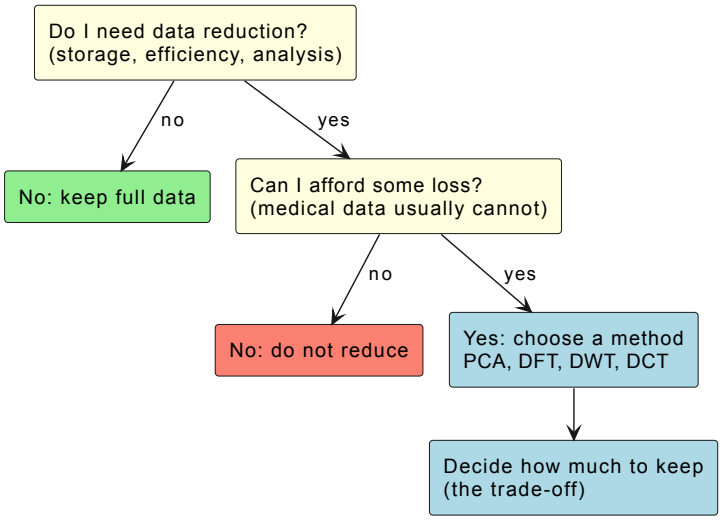
## 13. Should You Reduce Data at All

Data reduction is useful for **storage**, for **efficiency**, and for general-purpose **analysis** (when you simply want to look at the data in a simpler form). PCA and DFT are two such tools, and DWT and DCT are others.

But it cannot be a blind choice of "I need to reduce my data, so I'll apply PCA" or "somebody applies DFT". You must:

1. **First decide whether data reduction is necessary or beneficial** for your data, and in what form.
2. **Understand the implications** of reducing it (what you lose, and whether you can afford to lose it).
3. **Only then** choose a tool such as PCA or DFT.

In other words, establish the **context** first. The tools can wait.




### What comes next

With this we conclude the topic of data reduction. The next topic is **clustering**, an interesting aspect of data mining and a form of **unsupervised learning**, beginning with a very basic method called **k-means clustering**.

## 14. Practice Questions

Work these out on paper. They are designed to make you *use* the ideas, not recite them.

**1. Units matter.** Suppose the exam scores in the dataset were recorded out of $100$ instead of being normalized (every $x_2$ value multiplied by $100$), while hours stay the same. Without doing the full eigen-decomposition, predict what happens to the covariance matrix, to which variable dominates PC1, and to the "percentage of variance explained by PC1". What does this tell you about why we never mix hours and marks naively, and what would you do to the data *before* PCA to avoid the problem?

**2. Choosing how many components.** A dataset of $1000$ students has $4$ variables, and its covariance matrix has eigenvalues $9,\ 3,\ 1.5,\ 0.5$. (a) What percentage of the variance does each component explain? (b) How many components must you keep to retain at least $90\%$? (c) How many numbers do you store before and after reduction, ignoring $\mu$ and the eigenvectors? (d) Why does the second answer *not* depend on the number of students?

**3. Where does the error go?** In the lecture example, project each centered data point onto $v_2 = \frac{1}{\sqrt{2}}(-1, 1)$ to get $Z_2 = X_c v_2$. Which point has the unusually large value, and how does this connect to the large reconstruction error you saw for that same point? Then verify that $\frac{1}{N-1}\sum z_{2,i}^2$ equals $\lambda_2$, and explain in a sentence why this must be true.

**4. The hospital proposal.** A hospital data team proposes applying PCA to patient vital-sign records, keeping enough components to retain $98\%$ of the variance, to cut storage costs by $60\%$. Argue **for** and **against** the proposal. What would you want to know about the data and its use *before* deciding, and what could be hiding inside the discarded $2\%$?

**5. Same total, different story.** Dataset A has eigenvalues $(6.9,\ 0.1)$ and dataset B has eigenvalues $(3.6,\ 3.4)$. Both have total variation $7.0$. (a) Which dataset compresses better with one component, and why? (b) What would the off-diagonal covariance entry look like for each (large or near zero)? (c) Make the parallel to DFT: what kind of signal would compress well with just $X(0), X(1)$, and what kind of signal would not?